

5. A company manufactures a product at 4 manufacturing plants and sells it at 5 warehouses. The following table lists the capacity at each plant, the demand at each warehouse and the profit obtained per unit product by transporting the product from the plant to the warehouse and selling there.

Warehouses	Plants				Demand at Warehouse
	1	2	3	4	
A	6	6	11	15	80
B	4	6	10	12	120
C	6	4	7	6	150
D	4	10	14	14	70
E	8	8	7	9	90
Supply at Plant	100	200	120	80	

600

Based on the above data, answer the following questions

[Marks 6]

- Find the initial solution using Vogel's Approximation Method (VAM)
- Is the solution found using VAM optimal. If yes, justify your answer. If no, find the optimal solution.
- Identify which warehouses are unable to fulfill their demand.

not

**** End of the question paper ****

- 4 (a) Five wagons are available at stations 1, 2, 3, 4 and 5. These are required at five stations A, B, C, D and E. The mileages between various stations are given by the table below. How should the wagons be transported so as to minimize the total mileage covered?

	A	B	C	D	E
1	11	17	8	16	20
2	9	7	12	6	15
3	13	16	15	12	16
4	21	24	17	28	26
5	14	10	12	11	13

- 4(b) Solve the following game problem by using the principle of dominance and find the game value.

		Player B			
		I	II	III	IV
Player A	1	3	2	4	0
	2	3	4	2	4
	3	4	2	4	0
	4	0	4	0	8

- 5(a) Consider the following payoff matrix for two firms to advertise their products. Find graphically the best mixed strategy for both the firms and also find the value of the game.

		Firm II			
Advertising Method	Print	Television	Podcast	Radio	
Firm I	Print	19	6	7	5
	Television	7	3	14	6
	Podcast	12	8	18	4
	Radio	8	7	13	1

- 5(b) Solve the following NLP

$$\text{Optimize } Z = 2x_1^2 + x_2^2 + 3x_3^2 + 10x_1 + 8x_2 + 6x_3 - 100$$
Subject to $x_1 + x_2 + x_3 = 20$, $x_1, x_2, x_3 \geq 0$

*****End*****